

Week 3 Progress Report

Project: Log File Analyzer for Intrusion Detection

1. Introduction

In Week 3, the project was enhanced by adding visualization and reporting capabilities. The goal was to present detected threats in a clear and interpretable manner, enabling easier analysis of suspicious activities.

2. Objectives Achieved

- 1 Implemented data visualization using matplotlib
- 2 Generated graphical representation of top IP addresses
- 3 Developed report generation system (TXT and CSV)
- 4 Improved output formatting with severity levels

3. Technical Work

Visualization was implemented using matplotlib to display the most active IP addresses in a bar chart. The system generates a graph (graph.png) that highlights traffic distribution across IPs. Additionally, a reporting module was developed to export results into structured formats such as TXT and CSV. The TXT report summarizes detected threats, while the CSV file stores processed log data for further analysis. Output formatting was improved by introducing severity labels (HIGH, MEDIUM) based on activity levels.

4. Tools and Technologies Used

- 1 Python 3
- 2 matplotlib for visualization
- 3 pandas for data export
- 4 Kali Linux environment

5. Outcome

By the end of Week 3, the system was capable of not only detecting threats but also visualizing and reporting them effectively. The addition of graphical outputs and structured reports made the tool more user-friendly and suitable for real-world analysis.

6. Conclusion

Week 3 completed the core development of the project by integrating visualization and reporting features. The project now functions as a lightweight intrusion detection system with analysis, detection, and reporting capabilities.